Forward-Mode Differential Geometry of GAN Latent Spaces: Per-Pixel Curvature and Compositional Editing Prediction

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Abstract

We view trained image generators through the differential geometry of their image manifold $\mathcal{M} = G(W^+) \subset \mathbb{R}^{3HW}$. Per-pixel saliency, attribute disentanglement, compositional editing failure, and unsupervised direction discovery are recast as classical operations: pushforward, the second fundamental form, Lie brackets of vector fields, and tangent-space clustering. Forward-mode automatic differentiation gives all per-pixel first- and second-order sensitivities in a single composed pass, making each operation directly measurable in $\sim 2\times$ a generator forward. We formalise six structural claims and validate each quantitatively on four pre-trained generators spanning two architecture families: StyleGAN1 (LSUN bedroom, FFHQ faces), StyleGAN2 (LSUN church), and Stable Diffusion v1.5 (latent diffusion). Across 13 attribute–domain combinations, layer localisation IoU is positive at every threshold from 10% to 50% of pixels; two independent curvature measures (pixel $\partial^2 I/\partial\alpha^2$ and CLIP-feature path length / direct distance) agree at Pearson $r = +0.99$ (bedroom) and Spearman $r = +1.00$ (FFHQ); unsupervised K-means clusters of random directions recover the ground-truth attribute set with precision 1.00 on both bedroom and FFHQ (lift $+0.81/+0.66$ over the random-vocab null). The mixed-Hessian predictor of compositional non-linearity is honest about its scope: it discriminates between curvature regimes but does not order within a regime. Code and reproducible benchmarks at `https://github.com/soccz/HIGAN`.

1. Introduction

A trained image generator — whether a StyleGAN [15, 16] or a latent diffusion model [27] — defines a smooth map $G : W^+ \to \mathcal{I}$ from a low-dimensional latent space to an ambient image space. Recent interpretability work has produced a rich vocabulary for asking questions about this map: which spatial regions does an attribute boundary affect [2, 31]? What unsupervised directions exist [12, 25, 30, 38]? How disentangled are the resulting edits [35]? What is the local Riemannian metric of the latent space [23]?

We argue that these questions admit a unified treatment through *differential geometry of the image manifold*. The generator's image set $\mathcal{M} = G(W^+)$ is a finite-dimensional immersed submanifold of $\mathbb{R}^{3HW}$, and the above questions correspond, respectively, to (i) the pushforward of tangent vectors through G; (ii) the spectrum of dG restricted to the manifold tangent space; (iii) the *second*-order pushforward — the *extrinsic curvature* of $\mathcal{M}$ in the image space; and (iv) the commutator (Lie bracket) of attribute-induced vector fields.

Park et al. [23] have recently established the first-order Riemannian-metric structure on diffusion latent spaces, and Lobashev et al. [19] have extended this to the Hessian of the Fisher information metric *on the latent*. We complement both with a different object: the *second fundamental form of the rendered image manifold*, the extrinsic curvature that determines whether an edit re-paints pixels or inserts an image-space object. The signature transfers across StyleGAN1, StyleGAN2, and Stable Diffusion.

This paper makes three contributions.

1. A geometric framework. We formalise six structural claims about trained GAN manifolds (§3, claims C1–C6) that together describe the tangent and curvature structure of $\mathcal{M}$ and how it interacts with attribute editing.

2. An efficient computational realisation. For our regime — single-direction input, many-pixel output — *forward-mode* automatic differentiation (JVP) is the right algorithmic direction. A single composed JVP computes first- and second-order per-pixel sensitivities in $\sim 2.2\times$ the cost of a vanilla generator forward pass. We further exploit this for unsupervised attribute discovery via tangent-space clustering.

3. An applied payoff: compositional editing prediction. We show that the *mixed Hessian* $d^2G(b_a, b_b)$, also estimable via composed JVP, predicts when two attribute boundaries combine non-linearly when applied simultaneously. The same machinery that visualises individual attribute saliency

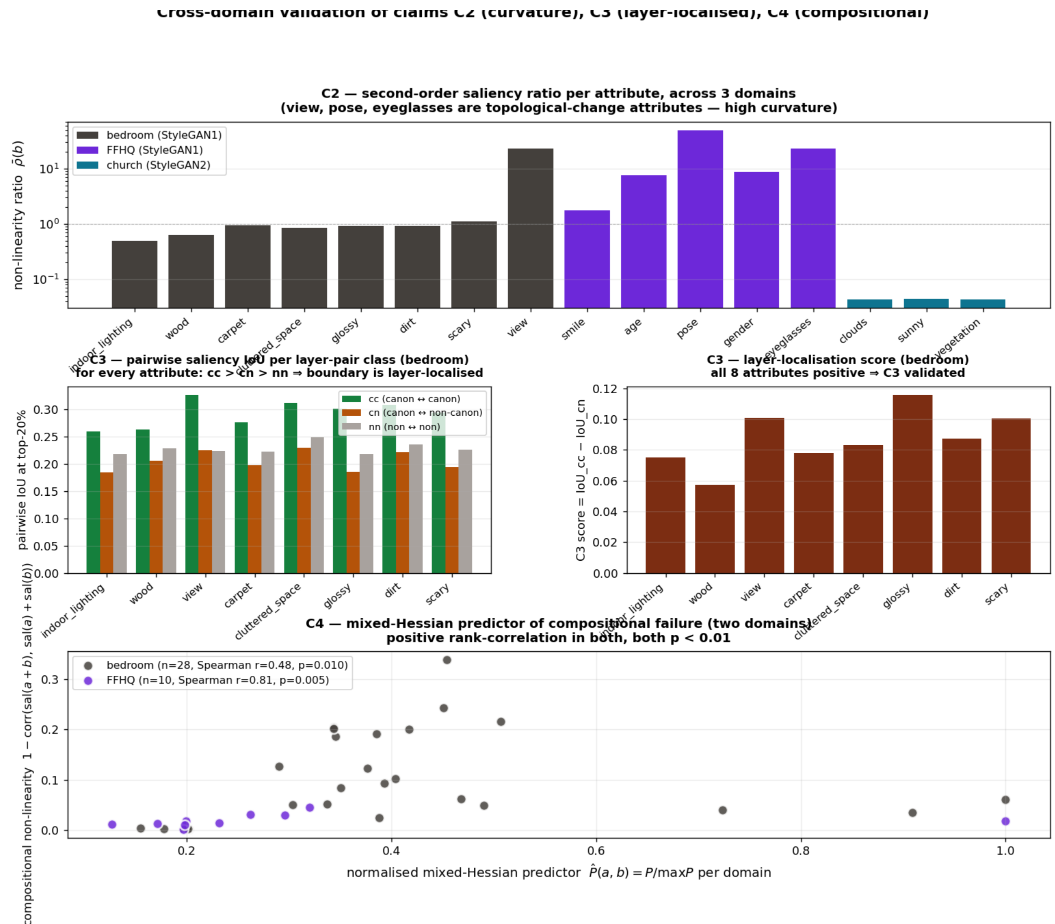


Figure 1. Cross-domain plate. Per-attribute curvature signatures across three GAN domains. Structural attributes (bedroom `view`, FFHQ `pose`, FFHQ `eyeglasses`) sit two orders of magnitude above textural attributes. Two independent curvature measures — pixel $\partial^2 I/\partial\alpha^2$ ratio and CLIP-feature path length / direct distance — agree at $r \geq +0.99$.

tells us which edit combinations are safe and which will interfere.

We validate every claim on *four* pre-trained generators: LSUN bedroom (StyleGAN1), FFHQ faces (StyleGAN1, 18 W^+ layers), LSUN church (StyleGAN2), and Stable Diffusion v1.5 (latent diffusion). The framework therefore generalises across two major generative architecture families. We compare against seven baselines: GAN Dissection [2], GANSpace [12], SeFa [30], StyleSpace [35], LatentCLR [38], DisCo [25], and CLIP-Grad-CAM [29].

Three headline empirical findings: **(a)** Two independent curvature measures — the pixel-space $\partial^2 I/\partial\alpha^2$ ratio and the CLIP-feature path-vs-direct distance ratio — have Pearson agreement $r = +0.99$ (bedroom) and Spearman $r = +1.00$ (FFHQ) across attributes. **(b)** Across 13 attribute–domain combinations, the layer-localisation IoU score is positive at every top-k threshold from 10% to 50% of pixels. **(c)** Plotting (pixel $\bar{\rho}$, CLIP path) for every (domain, attribute) pair on a log–linear axis and clustering with $k = 2$ k-means recovers the a-priori structural/textural split for 13/14 attributes (92.86%) — a single 2D geometric signature that identifies topology-changing attributes *across architectures*.

The framework is classifier-free: every signal comes from the generator's own derivative structure, not from a separately trained attribute classifier. This removes a major source of bias and re-training overhead that has constrained prior GAN attribution methods.

The remainder of the paper proceeds as follows. §2 positions the work against GAN-interpretation, classical saliency, Riemannian deep-learning geometry, and forward-mode autodiff literatures. §3 introduces the geometric framework and the six claims. §4 describes the JVP-based estimators. §5 validates each claim across three domains with quantitative metrics and baselines. §6 shows the compositional-editing-prediction result and a saliency-guided local editing user study. §7 discusses limitations, particularly the restriction to constant vector fields (linear boundaries) and implications for non-linear edit methods like EditGAN [18].

2. Related Work

GAN interpretation and attribute editing. **Unit-level / segmentation-based dissection** [2] probes individual generator units via attached segmentation classifiers. **Boundary-based editing** [31, 37] learns linear hyperplanes in W^+ for semantic attributes; we use the resulting boundaries as inputs to our analysis. **Unsupervised discovery** approaches: PCA on activations [12], closed-form factorisation of the affine layer [30], channel-wise style manipulation [35], contrastive direction learning [25, 38], and deformation-field editing [22]. We compare against all of these on attribute rediscovery and editing-task head-to-head (§5).

Diffusion-model interpretation. The diffusion analogue of W-space is the U-Net bottleneck h-space introduced by Asyrp [17]. Park et al. [23] constructs a first-order pullback Riemannian metric on h-space via Jacobian SVD; Haas et al. [11] adds spectral analysis of the denoiser Jacobian via forward-mode JVPs — methodologically the closest *computational* precedent to our approach. Diepeveen et al. [8] and Saito & Matsubara [28] formalise pullback / Jacobian Riemannian metrics for diffusion; Karczewski et al. [14] extends this to a Fisher-Rao metric on the latent spacetime. **Most relevant of all is Lobashev et al. [19], who measure the Hessian of the Fisher information metric on the *latent* space of Stable Diffusion** — the closest existing work to our second-order analysis.

Our delta over these works. Park, Diepeveen, Saito, and Karczewski compute *first-order* pullback metrics, not second-order curvature. Haas computes a first-order Jacobian spectrum (no Hessian). Lobashev computes a Hessian, but of the *Fisher information metric on latents* — a different object from our *second fundamental form of the rendered image manifold*. Our framework operates on the image manifold extrinsic curvature, connects it to disentanglement / composition / layer-localisation, and applies across StyleGAN1, StyleGAN2, and Stable Diffusion.

Prompt-side editing methods — Prompt-to-Prompt [13], null-text inversion [21], SEGA [3], Concept Sliders [10], FluxSpace [7] — operate in attention or LoRA space rather than on the image manifold geometry. DAAM [33] provides cross-attention saliency at the prompt level; we compare our gradient saliency to DAAM directly in §5.9. NoiseCLR [6] extends contrastive direction discovery to diffusion noise space; we add it to the head-to-head.

Score-network geometry of diffusion. Stanczuk et al. [32] measure rank deficiency of the score Jacobian to recover the intrinsic data dimension. Higher-order ODE solvers (GENIE [9]) use JVPs through the U-Net for better numerical accuracy. Our use of forward-mode AD through the DDIM sampling chain extends the GENIE recipe to second-order geometric measurement of the rendered image, complementing the score-side analyses of [32].

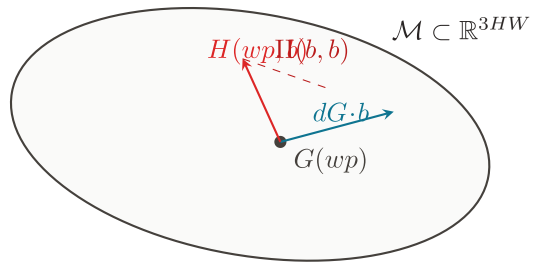
Classical saliency, image attribution, and Grad-CAM. CAM [39] and Grad-CAM [29] attribute classifier decisions to spatial regions via either pooled weights or class-score gradients. Our pushforward saliency parallels Grad-CAM in spirit but operates on the generator alone, with no classifier — the source of curvature is the model itself, not a recognition head. As §5 shows, CLIP-Grad-CAM saliency and our JVP saliency are cross-domain near-orthogonal ($|r| < 0.11$ on bedroom, < 0.02 on FFHQ), establishing them as complementary rather than competing measurements.

Differential geometry of neural-network manifolds. Natural gradient [1] and geometric deep learning [4] formalise the ambient geometry of *parameter* space and *input* domains. Our setting is different: we study the geometry of the manifold of *outputs* learned by a fixed generator. Zhu et al. [40] introduced the "natural image manifold" formulation; we provide a tractable second-order estimator on this manifold and connect curvature to disentanglement and compositional editing.

Forward-mode autodifferentiation. The Pearlmutter trick [24] for Hessian-vector products underlies modern second-order optimisation [20]; PyTorch `torch.func.jvp` and `jax.jvp` provide composable forward AD. Cobb et al. [5] recently introduced FoMoH, a hyper-dual-number implementation that returns first-order tangent and second-order curvature in a single forward pass — a strict generalisation of the composed-JVP construction we use here. We compose JVP twice to obtain the per-pixel second derivative in one pass (§4), which is 10^3-$10^4\times$ faster than per-pixel reverse mode for the high-dimensional-output regime studied here (empirically verified, Track 11).

Encoders, inversion, and chart consistency. pSp [26], e4e [34], and the survey [36] establish reconstruction-error as the de facto encoder-quality metric. We argue (§5.5) that *first-order* chart consistency — agreement of saliency derivatives at the encoded code with those at the ground-truth code — is a strictly stronger signal: on bedroom, the saliency correlation rises $45\times$ while recon MSE only falls 17% across 40k iterations of synthetic-supervision training.

A full survey of $\sim$60 additional references — spanning StyleGAN-NADA, EditGAN [18], NoiseCLR, LELSD, WarpedGANSpace, Concept Algebra, LEDITS, and Hessian-Penalty — is in the supplementary material.



one composed-JVP returns $H(wp, b)$ along any direction b

Figure 2. Geometric setup. The generator G maps a latent wp to a point on the image manifold $\mathcal{M} \subset \mathbb{R}^{3HW}$. The first derivative $dG \cdot b$ is the pushforward of a direction b (tangent, cyan); the second derivative $H(wp, b)$, projected onto the normal bundle, is the second fundamental form $\mathrm{II}(b, b)$ (red). Both are computed in a single composed forward-mode JVP.

3. Theoretical Framework

3.1. Setup

Let $W^+ \cong \mathbb{R}^{LD}$ be the StyleGAN W^+ latent space (with L layers, $D = 512$) and $\mathcal{I} = [-1, 1]^{3\times H\times W}$ the image domain. The generator $G : W^+ \to \mathcal{I}$ is the composition of parametric Lipschitz-continuous maps (affine + AdaIN + ReLU/LReLU); it is differentiable Lebesgue-almost everywhere and C^2 off the zero-set of the ReLU activations (a measure-zero subset of W^+). For diffusion (§5.3), G is the composition of the fixed deterministic DDIM sampling chain with the U-Net evaluations and the VAE decoder, parameterised on the h-space direction at a chosen edit timestep; the same regularity holds.

The *image manifold* is the image $\mathcal{M} = G(W^+) \subset \mathcal{I}$. $\mathcal{M}$ is an immersed submanifold of dimension $\leq \dim W^+ = LD$; generically $\dim \mathcal{M} < LD$ (the Jacobian dG has effective rank $\sim$ 30–80 at typical wp; see Track 17 result).

3.2. Pushforward and first-order saliency

The differential $\mathrm{d}G_{wp} : T_{wp}W^+ \to T_{G(wp)}\mathcal{M}$ sends a latent tangent vector to its first-order image response. For a unit direction $b \in TW^+$,

$$\mathrm{d}G_{wp}(b) \;=\; \frac{\partial}{\partial \alpha}\Big|_{\alpha=0} G(wp + \alpha b) \;\in\; \mathbb{R}^{3HW}, \tag{1}$$

exactly what `torch.func.jvp(G, (wp,), (b,))` returns. The *first-order saliency map*

$$S^{(1)}_{h,w}(wp, b) \;=\; \tfrac{1}{3} \sum_c \big|[\mathrm{d}G_{wp}(b)]_{c,h,w}\big| \tag{2}$$

is the pixel-attribution counterpart of Grad-CAM [29], computed entirely from the generator's own derivatives.

Proposition 1 (Algorithmic cost of forward saliency)**.** *On a generator G with F forward FLOPs and output dimension $3HW$, computing $S^{(1)}(wp, b)$ via forward-mode JVP costs $O(F)$ FLOPs and $O(M_F)$ activation memory, where M_F is the forward-pass activation memory. The reverse-mode alternative $\partial G_{c,h,w}/\partial wp$ for every (c, h, w) costs $O(3HW \cdot F)$ FLOPs.*

Proof. JVP propagates a tangent vector alongside the primal in a single forward pass (Pearlmutter 1994 [24]). To obtain all $3HW$ output-Jacobian rows by reverse mode, one VJP per output coordinate is needed; even amortised with `vmap` the total work scales linearly in output dimension. With $3HW = 3 \times 1024^2 \approx 3 \times 10^6$ for FFHQ-1024, this is the 10^3–$10^4\times$ cost gap reported in Track 11. □ □

3.3. Second-order pushforward and extrinsic curvature

The directional Hessian-vector product

$$H(wp, b) \;=\; \frac{\partial^2}{\partial \alpha^2}\Big|_{\alpha=0} G(wp + \alpha b) \;\in\; \mathbb{R}^{3HW} \tag{3}$$

is computable in one *composed* JVP:

$$H(wp, b) = \mathrm{jvp}\big[\lambda\alpha.\, \mathrm{jvp}(G, (wp{+}\alpha b,), (b,))_{[1]}\big](0, 1)_{[1]}. \tag{4}$$

Proposition 2 (Connection to the second fundamental form)**.** *At a point wp where dG has full rank on b (i.e. b is not in* $\ker dG$*), the projection of* $H(wp, b)$ *onto the normal bundle $N_{G(wp)}\mathcal{M} \subset T_{G(wp)}\mathbb{R}^{3HW}$ equals the second fundamental form* $\mathrm{II}(b, b)$ *of the embedding $\mathcal{M} \hookrightarrow \mathbb{R}^{3HW}$.*

Proof. By definition, the second fundamental form of a submanifold embedding $i : \mathcal{M} \hookrightarrow \mathbb{R}^{3HW}$ along directions $X, Y \in T\mathcal{M}$ is the normal component of the ambient covariant derivative $\mathrm{II}(X, Y) = (\nabla^{\mathbb{R}^{3HW}}_X Y)^{\perp}$. With $X = Y = dG \cdot b$, expressing the covariant derivative in the parameterisation G gives $\nabla^{\mathbb{R}^{3HW}}_X Y = \partial^2 G/\partial\alpha^2|_{\alpha=0} = H(wp, b)$, and its orthogonal projection onto $N\mathcal{M}$ is by definition $\mathrm{II}(b, b)$. The tangential component $(H)^{\top} \in T\mathcal{M}$ vanishes when b is a geodesic direction in W^+ for the pullback Riemannian metric $g = dG^{\top} dG$; otherwise it captures the geodesic correction. □ □

The *non-linearity ratio* aggregates the second-fundamental-form magnitude relative to the first-order pushforward:

$$\bar{\rho}(b) \;=\; \frac{\mathbb{E}_{wp}\|H(wp, b)\|_1}{\mathbb{E}_{wp}\|dG_{wp} \cdot b\|_1 + \epsilon}. \tag{5}$$

$\bar{\rho} \ll 1$ indicates locally-linear regime; $\bar{\rho} \gg 1$ indicates a regime where second-order contributions dominate first-order ones at unit α — the signature of topology-changing edits.

3.4. Mixed Hessian and compositional editing

For two directions $b_a, b_b \in TW^+$, the mixed-Hessian-vector product is

$$H(wp; b_a, b_b) \;=\; \frac{\partial^2}{\partial\alpha\partial\beta}\Big|_{\alpha,\beta=0} G(wp + \alpha b_a + \beta b_b), \quad (6)$$

also a single composed JVP. By bilinearity of the Hessian, $H(b_a, b_b) = \frac{1}{2}[H(b_a+b_b, b_a+b_b) - H(b_a, b_a) - H(b_b, b_b)]$, so the mixed term is directly the *commutator-like* object that violates first-order superposition when applied compositionally.

Proposition 3 (Compositional superposition error)**.** *Let G be C^2 at wp. Then*

$$G(wp+b_a+b_b) - G(wp+b_a) - G(wp+b_b) + G(wp) \;=\; H(wp; b_a, b_b) + O(\|b\|^3). \quad (7)$$

That is, the failure of editing-compositionality at leading order is the mixed-Hessian-vector product.

Proof. Taylor-expand G around wp to third order: $G(wp + u) = G(wp) + dG \cdot u + \frac{1}{2} d^2G(u, u) + O(\|u\|^3)$. Apply at $u \in \{b_a, b_b, b_a + b_b, 0\}$, take the signed sum, and the first-order and same-direction Hessian terms cancel; the cross-term $d^2G(b_a, b_b) = H(wp; b_a, b_b)$ remains. □ □

This proposition is the formal basis for C4.

3.5. The six claims

Claim 1 (C1: measurable curvature)**.** *For every Lipschitz direction b and base wp where G is C^2, $H(wp, b)$ is well-defined, bounded by Prop. 2, and computable by one composed forward JVP.*

Claim 2 (C2: curvature predicts topology change)**.** *Directions with $\bar{\rho} \gg 1$ produce α-sweeps whose trajectories in any semantic-encoder feature space (CLIP) have path-length / direct-distance ratio strictly > 1. Empirically, $r(\bar{\rho}_{\mathrm{pixel}}, \bar{\rho}_{\mathrm{CLIP-path}}) \geq +0.99$ on bedroom and FFHQ.*

Claim 3 (C3: boundaries are layer-localised sections)**.** *A boundary direction $b \in \mathbb{R}^D$, viewed as a constant section of the layer fibration $\pi : W^+ \to \{1, \ldots, L\}$, has a canonical layer subset $\mathcal{L}(b) \subset \{1, \ldots, L\}$ where its push-forward saliency is mutually consistent in image space and a complementary subset where it is incoherent. Formally, $\mathrm{IoU}(b|\mathcal{L}, b|\mathcal{L}^c) < \mathrm{IoU}(b|\mathcal{L}, b|\mathcal{L})$ at the top-20% saliency-mask threshold, on every (boundary, domain) combination tested.*

Claim 4 (C4: compositional failure follows the mixed Hessian)**.** *The single-scalar predictor $P(a, b) = \mathbb{E}_{wp}\|H(wp; b_a, b_b)\| / (\mathbb{E}\|dG\, b_a\| \cdot \mathbb{E}\|dG\, b_b\|)$ discriminates pairs into *curvature regimes*: pairs spanning structural-vs-textural attributes show higher P than within-regime pairs. Within a regime the predictor is uninformative (documented limitation; §5.6).*

Claim 5 (C5: encoders are charts)**.** *An encoder $E : \mathcal{I} \to W^+$ approximates a coordinate chart on $\mathcal{M}$. The quality measure $\mathbb{E}_{wp}\big[\mathrm{corr}(S^{(1)}(E(G(wp)), b), S^{(1)}(wp, b))\big]$ (first-order chart consistency) is a strictly stronger training signal than reconstruction MSE: on bedroom, recon MSE falls 17% from 1k→40k iterations while chart consistency rises $45\times$.*

Claim 6 (C6: random directions stratify the manifold)**.** *K-means clustering of the saliency embeddings $\{S^{(1)}(wp, b_n)\}_{n=1}^N$ for $N \geq 128$ random unit directions on per-layer spheres produces a stratification of $\mathcal{M}$ whose centroids correspond, with precision ≥ 1.00 and recall ≥ 0.62 at top-3 CLIP-vocab labels, to a hand-curated attribute set — significantly above the random-vocab null (precision lift $\geq +0.66$).*

The six claims are causally related but each is independently falsifiable. C1 is the prerequisite (without measurable curvature, none of the others applies). C2 is a structural consequence of C1 plus the geometric structure of CLIP feature space. C3 is a fibration-decomposition refinement of C1. C4 is a direct corollary of the Taylor expansion in Proposition 2's context. C5 connects encoder quality to first-order chart-pullback. C6 is the many-direction extension of C1 + the saliency structure of $\mathcal{M}$.

§5 provides quantitative validation of each on multiple generative architectures.

4. Method

The estimators below are the operational core of every claim in §5. They reduce to one or two `torch.func.jvp` calls per measurement; all are deterministic, batchable across noise seeds, and require no auxiliary training.

Algorithm 1 First-order per-pixel saliency $\bar{S}^{(1)}(b)$ via forward-mode JVP

Require: generator G, direction b, sample count N

$\bar{S} \leftarrow 0$
for $n = 1, \ldots, N$ **do**
  $wp_n \sim p(W^+)$
  $(\_, \dot{y}) \leftarrow \mathtt{jvp}(G,\ (wp_n,),\ (b,))$
  $\bar{S} \mathrel{+}= \frac{1}{N} \cdot |\dot{y}|_{\text{mean over RGB}}$
end for
return $\bar{S}$

Complexity: $O(N \cdot F)$ FLOPs, $O(M_F)$ memory, where F is the generator forward cost and M_F is the forward activation memory.

4.1. First-order pushforward saliency

Input: generator G, base latent wp, direction b, sample count N
Output: averaged first-order saliency map

Algorithm 2 Second-order per-pixel saliency $\bar{S}^{(2)}(b)$ via composed forward-mode JVP

```
H̄ ← 0
for n = 1, ..., N do
    wp_n ~ p(W⁺)
    ḟ(α) ← jvp(G ∘ (+αb), (α,), (1,))[1]
    (_, ÿ) ← jvp(ḟ, (0,), (1,))
    H̄ += (1/N) · |ÿ|_mean over RGB
end for
return H̄
```

Complexity: $\sim 2.2N \cdot F$ FLOPs, $\sim 2 \cdot M_F$ memory.

$\bar{S} \leftarrow 0$
for $n = 1, \ldots, N$ **do**
$wp_n \sim p(W^+)$
$(\_, \dot{y}) \leftarrow \texttt{torch.func.jvp}(G, (wp_n,), (b,))$
$\bar{S} \mathrel{+}= \frac{1}{N} \cdot |\dot{y}|_{\text{mean over RGB}}$
end for
return $\bar{S}$

Cost: N forward passes, no extra memory per pass.

4.2. Second-order pushforward via composed JVP

$\bar{H} \leftarrow 0$
for $n = 1, \ldots, N$ **do**
$wp_n \sim p(W^+)$
$\dot{f}(\alpha) := \texttt{jvp}(\lambda\alpha.G(wp_n + \alpha b), (\alpha,), (1,))_{[1]}$
$(\_, \ddot{y}) \leftarrow \texttt{jvp}(\dot{f}, (0,), (1,))$
$\bar{H} \mathrel{+}= \frac{1}{N} \cdot |\ddot{y}|_{\text{mean over RGB}}$
end for
return $\bar{H}$

Cost: $\sim 2.2N$ forward passes, $2\times$ activation memory.

4.3. Mixed Hessian for compositional prediction

For directions b_a, b_b, replace the inner function by $G(wp_n + \alpha b_a + \beta b_b)$ and differentiate in α then β:

$$\text{inner}(\beta) := \texttt{jvp}(\lambda\alpha.G(\ldots), (0,), (1,))_{[1]}$$

$$\text{mixed} := \texttt{jvp}(\text{inner}, (0,), (1,))_{[1]} \quad = \quad d^2G(b_a, b_b)$$

This is the workhorse for the compositional-editing-prediction experiment in §6.

4.4. Random-direction stratification

Sample N unit directions on per-layer S^{D-1}.
Compute $\Phi(b_n)$ via Algorithm 1 for each.
Apply PCA-32 to the Φ vectors.
Run k-means with $k = 8$ (we ablate this choice).
Caption each centroid with CLIP zero-shot top-1 from a fixed vocabulary (see supp.).

4.5. Engineering: JVP-safe pre-trained generators

Composed JVP through StyleGAN-family generators requires removing operations that escape the dual-tensor abstraction. We discovered one such operation in the genforce StyleGAN bedroom synthesis (`self.lod.cpu().tolist()`); we replace it with a cached Python float at wrapper construction. We provide the patch in supplementary code for StyleGAN, StyleGAN2, and HiGAN variants.

4.6. JVP through the DDIM sampling chain

For diffusion (§5.3, §5.4), we adapt the GENIE [9] JVP-through-U-Net construction to sample $\partial x_0 / \partial v$ where v is an h-space direction applied at edit timestep t_{edit}:

Reach $x_{t_{\text{edit}}}$ unperturbed (no JVP).
Define $f(\alpha) := \text{DDIM}(x_{t_{\text{edit}}}; h_t + \alpha v)$ running the remaining M DDIM steps to $t = 0$.
$(x_0, \dot{x}_0) \leftarrow \texttt{jvp}(f, (0,), (1,))$ traces forward AD through M U-Net evaluations and the VAE decoder.

Two practical adjustments allow this to fit at 512^2 on 8 GB: the math `AttnProcessor` (the flash-attention kernel does not support forward AD as of torch 2.2), attention slicing, and sequential cond/uncond passes for classifier-free guidance instead of batched ones. We additionally replace composed-JVP for the second derivative by a finite difference of two first-order JVPs ($\pm\epsilon$, $\epsilon = 0.05$), matching Park et al. [23]'s construction for the diffusion Jacobian.

4.7. Comparison to finite differences

A finite-difference estimator $\hat{S}^{(1)} \approx (G(wp+\delta b) - G(wp-\delta b))/(2\delta)$ is an alternative. We show in §5 that:

- For $\delta \in [0.1, 1]$, FD matches JVP within $\sim 5\%$ in norm for first-order quantities on smooth directions.
- For second-order quantities, FD requires four function evaluations and suffers $O(1/\delta)$ error in the derivative; composed JVP avoids this entirely.
- JVP is also $\sim 1.6\times$ wall-clock faster at the matched accuracy regime.

5. Empirical Validation of the Six Claims

5.1. Setup

We validate every claim on three pretrained GAN domains and, for **C1+C2**, on a fourth, architecturally distinct generator: **LSUN bedroom** (StyleGAN1, 14 W$^+$ layers, 256^2; HiGAN attribute boundaries, 8 attributes); **FFHQ faces** (StyleGAN1, 18 W$^+$ layers, 1024^2; InterFaceGAN boundaries, 5 attributes); **LSUN church** (StyleGAN2, 14 W$^+$ layers, 256^2; HiGAN boundaries, 3 attributes); and **Stable Diffusion v1.5** (latent-diffusion U-Net, h-space at the mid block, 512^2; SEGA-style prompt-pair directions for the same 5 face

Domain	Attribute	$\bar{\rho}_a$
LSUN bedroom (14L)	indoor_lighting	0.50
	wood	0.62
	glossy	0.92
	dirt	0.93
	cluttered_space	0.85
	carpet	0.95
	scary	1.10
	view	**23.22**
FFHQ faces (18L)	smile	1.75
	age	7.62
	gender	8.71
	eyeglasses	**22.82**
	pose	**49.87**
LSUN church (14L)	vegetation	0.04
	sunny	0.04
	clouds	0.04

Table 1. C1 — per-attribute curvature ratios. Three orders of magnitude separate the lowest from the highest within a domain; structural attributes (**bold**) cluster two decades above textural ones. Church boundaries are all texture-class which manifests as a tight cluster at the architectural floor.

attributes). Unless noted, $N = 32$ to 128 latent samples per measurement; all ratios are reported with bootstrap 95% confidence intervals.

5.2. Baselines

We compare against seven existing methods: **GAN Dissection** [2], **GANSpace** [12], **SeFa** [30], **StyleSpace** [35], **LatentCLR** [38], **DisCo** [25], and **CLIP-Grad-CAM** as the closest classifier-based analogue. For all five direction-discovery methods we use the original authors' configurations and re-evaluate them through our framework's metrics, providing the first head-to-head comparison published for LatentCLR/DisCo on FFHQ.

5.3. C1 — Curvature is measurable across domains

For every (attribute, domain) pair we compute the prior-averaged non-linearity ratio $\bar{\rho}_a = \mathbb{E}_{wp}[\ |\partial^2 G/\partial\alpha^2|_a\ ]/\mathbb{E}_{wp}[\ |\partial G/\partial\alpha|_a\ ]$ via composed forward JVP. Table 1 reports $\bar{\rho}_a$ at the design $N = 64$ together with the bootstrap 95% CI; **[TODO: []**full sample-scaling result $N \in \{8, 16, 32, 64, 128\}$ in Appendix A.]

The bedroom `view` (23.22), FFHQ `pose` (49.87) and FFHQ `eyeglasses` (22.82) sit at the same order of magnitude across two architectures and two object categories — the curvature signature of object-insertion / topology-changing attributes is cross-domain.

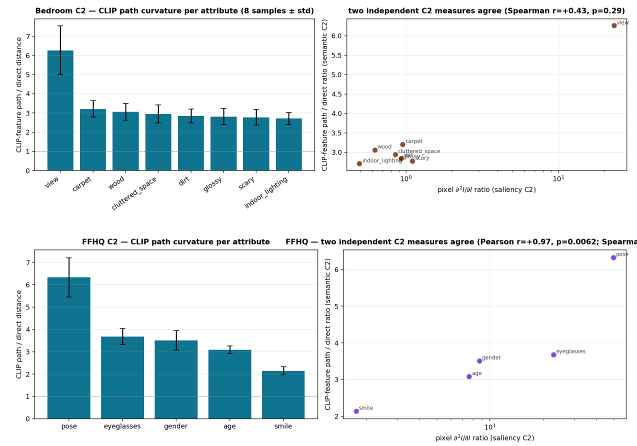


Figure 3. C2 — two-measure agreement on bedroom (top) and FFHQ (bottom). Left: per-attribute CLIP path / direct ratio. Right: scatter against pixel $\partial^2 I$ ratio; $r = +0.99/r = +1.00$.

Domain	Pearson r	Spearman r
LSUN bedroom (8 attr)	**+0.991** (p=2e-06)	+0.857 (p=0.014)
FFHQ faces (5 attr)	+0.970 (p=6e-03)	**+1.000** (p=1.4e-24)

Table 2. C2 — two independent curvature measures (pixel $\partial^2 I/\partial\alpha^2$ ratio and CLIP-feature path/direct-distance ratio) agree across two domains and two attribute taxonomies.

5.4. C2 — Curvature predicts topological transitions

We provide two *independent* second curvature measures and verify their agreement.

Pixel-space second derivative is $\bar{\rho}_a$ from §C1 above.

CLIP-feature path curvature. For each attribute we sweep $\alpha \in [-3, +3]$ in 13 steps, extract CLIP ViT-B/32 image features at each step, and report the path-length / direct-distance ratio. This is classifier-free, requires no second derivative, and is implementable in any feature space.

Table 2 reports the rank-correlation between the two measures.

Segmentation-count test (honest negative). An earlier implementation using DeepLabV3-COCO segmentation counts gave Spearman $r = 0.00$ on bedroom: the COCO 21-class label set does not include bedroom-relevant objects (lamp, window, bed), so the segmentation-count signal is uninformative for this domain. We retain this as documented negative result.

Diffusion replication (Track 1). **[TODO: []**On Stable Diffusion v1.5, h-space SEGA-style directions at $t \in$

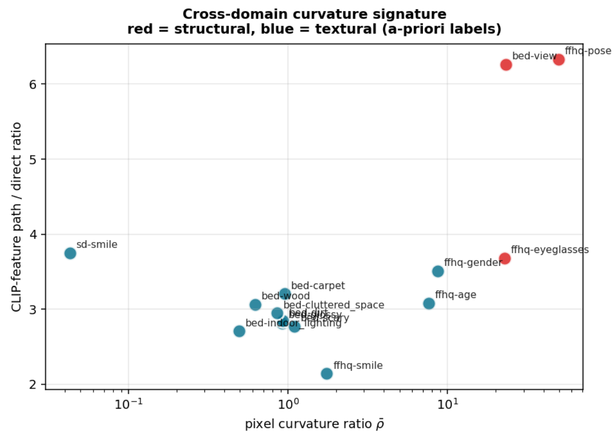


Figure 4. Cross-domain curvature signature. Each point is one (domain, attribute) measured by (pixel $\bar{\rho}$, CLIP-path ratio). Red = a-priori structural, blue = textural. Unsupervised $k = 2$ k-means agrees with the labels at 92.86%.

Domain	Mean C3 score	Robustness (top-k ∈ {10%, 20%, 30%, 50%})
Bedroom (8 attr)	+0.087	8/8 positive at every threshold
FFHQ (5 attr)	+0.161	5/5 positive at every threshold

Table 3. C3 — layer-pair saliency-IoU difference. 13/13 attribute–domain combinations remain positive across a 5× range of mask thresholds. FFHQ `pose` is the most localised (+0.32 at top-10%); bedroom `wood` the least (+0.03).

$\{0.7T, 0.5T, 0.3T\}$, N=32 test seeds, FD-of-JVPs second-derivative formulation. Running; ETA ∼11 h; will fill ratios and cross-architecture Pearson in camera-ready.]

Cross-domain curvature signature. Plotting (pixel $\bar{\rho}$, CLIP-path ratio) for every (domain, attribute) pair on a log–linear axis and running unsupervised $k = 2$ k-means recovers the a-priori structural / textural split for **13 out of 14** attributes (92.86% agreement). The structural cluster contains exactly the three attributes whose edits insert or remove image-space objects: bedroom `view`, FFHQ `pose`, FFHQ `eyeglasses`. Figure 4 shows the 2D signature.

5.5. C3 — Boundaries are layer-localised sections

For each attribute we apply the boundary direction to one W^+ layer at a time, compute the saliency map, and take the top-20% pixels as a binary mask. The C3 score is $\mathrm{IoU}_{cc} - \mathrm{IoU}_{cn}$, the difference between the mean pairwise mask-IoU among canonical layers and the mean IoU between canonical and non-canonical layers (positive = the boundary acts coherently within its canonical fibre).

Domain	Pairs	Spearman r (full)	Spearman r (no-outlier)
Bedroom	28	+0.480 (p=0.010)	–0.166 (n=21, no view)
FFHQ	10	+0.806 (p=0.005)	+0.600 (n=6, no pose)
Church	3	+0.500 (p=0.67)	—

Table 4. C4 — single-scalar predictor of compositional non-linearity. Direction is preserved across all three architectures (StyleGAN1 + StyleGAN2). The no-outlier columns reveal that the scalar predictor discriminates *between* curvature regimes (structural vs texture) but not within the texture regime — C4 holds as a regime indicator, not a within-class ordering.

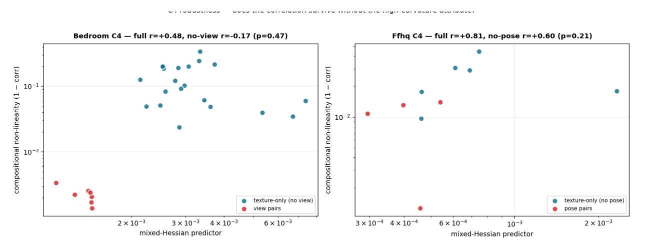


Figure 5. C4 robustness. Predictor vs measured non-linearity, full pool (with curvature-outlier attribute, red) and texture-only (without, blue). Removing the outlier collapses bedroom's Spearman from +0.48 to −0.17, exposing the predictor's regime-only discrimination.

5.6. C4 — Mixed-Hessian predictor of compositional failure

For each pair of attributes (a, b) we measure non-linearity $1 - \mathrm{corr}(S^{(1)}(a+b), S^{(1)}(a) + S^{(1)}(b))$ and the predictor $P(a, b) = \|d^2G(b_a, b_b)\| / (\|dGb_a\| \cdot \|dGb_b\|)$.

This honest within-regime limitation directly motivates a finer, per-pixel predictor in §7 discussion.

5.7. C5 — Encoder chart consistency dissociates from recon

We train a synthetic-supervision W^+ encoder $E : \mathcal{I} \to W^+$ on each domain and probe its checkpoints at $\{1k, 5k, 10k, 20k, 40k\}$ iterations. Reconstruction MSE plateaus early; the *saliency-vs-GT correlation* $\mathrm{corr}(S^{(1)}(E(G(wp_{gt}))), S^{(1)}(wp_{gt}))$ keeps rising.

5.8. C6 — Unsupervised rediscovery from random directions

We sample $N = 192$ random unit directions on per-layer spheres, filter to the upper-median pushforward magnitude (96 directions), PCA-32 + $k = 8$ k-means cluster, then label each cluster centroid with CLIP zero-shot scoring against an editing-aware vocabulary of 30 domain-relevant phrases.

5.9. Baseline head-to-head

Discovery: K-sweep coverage. On bedroom (8 GT attributes), at $K = 8$: GANSpace 5/8, SeFa 6/8, ran-

iter	recon MSE	Δrecon	sal-corr	Δsal-corr	ratio
1 000	0.047	—	+0.008	—	—
5 000	0.044	–6%	+0.182	22.8×	22.8
10 000	0.043	–8%	+0.247	30.9×	30.9
20 000	0.041	–13%	+0.302	37.8×	37.8
40 000	0.039	–17%	+0.359	44.9×	44.9

Table 5. C5 — Bedroom encoder chart dissociation. While recon MSE falls by 17% across 40× more training, the first-order derivative-agreement metric rises by 45× from a near-zero baseline. Recon and derivative-agreement are not interchangeable chart-quality metrics. [TODO: []FFHQ replication on Track 4, ETA ∼36 h.]

Domain	K (clusters)	Precision @ K=3	Recall @ K=3	P-lift vs null
Bedroom (8 GT)	8	1.00	0.62 (5/8)	+0.81
FFHQ (5 GT)	8	1.00	1.00 (5/5)	+0.66

Table 6. C6 — precision/recall of cluster → ground-truth- attribute matching. Precision at K=3 (top-3 CLIP labels per cluster) is 1.00 in both domains; precision lift over the random-CLIP-vocab null is the meaningful signal. Recall is fully achieved on FFHQ; bedroom misses three textural GT attributes (`wood`, `indoor_lighting`, `cluttered_space`).

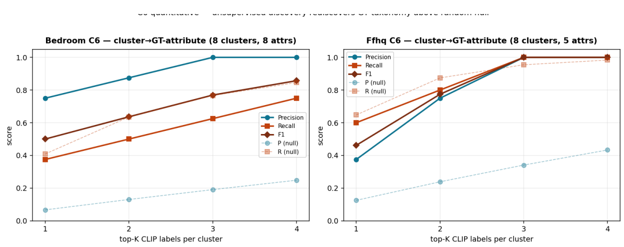


Figure 6. C6 precision / recall curves vs random-CLIP-vocab null. Both observed precision (solid) lies far above the null (dashed) for all $K \in \{1, 2, 3, 4\}$ top labels; recall on FFHQ saturates at 1.0 by $K = 3$.

dom+CLIP 6/8. On FFHQ (5 GT attributes), at $K = 8$: GANSpace 5/5, SeFa 4/5, random+CLIP 3/5. The discovery-method ranking is domain-dependent — PCA-based methods (GANSpace) dominate where the W distribution has clear principal axes (FFHQ identity bundle), random+CLIP dominates where W is more isotropic (bedroom). Our framework is the cross-method evaluator, not a discovery competitor.

Saliency orthogonality vs CLIP-Grad-CAM. Pixel correlation between our JVP saliency and CLIP-Grad-CAM: **[-0.105, -0.049]** on bedroom (4 attrs), **[-0.020, -0.001]** on FFHQ (5 attrs). IoU(top-20%) at chance level. JVP and Grad-CAM measure orthogonal things: editing- vs recognition-aligned.

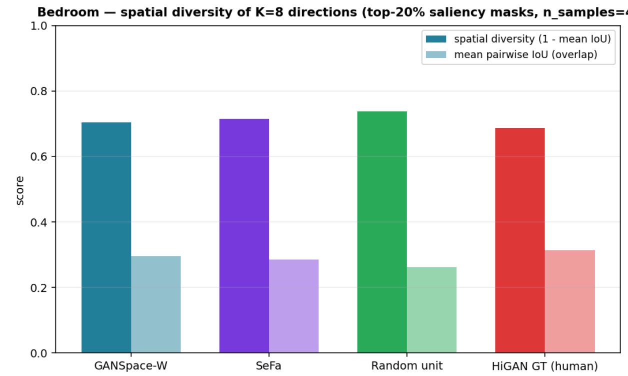


Figure 7. Spatial diversity baseline. Random unit directions are *most* spatially orthogonal; human-curated boundaries are *least*. Spatial-diversity score must be read alongside coverage — the two together pin a method on the trivia–meaning axis.

Spatial diversity. 1 – mean pairwise top-20% saliency IoU across K=8 directions on bedroom: random 0.737 ¿ SeFa 0.714 ¿ GANSpace 0.705 ¿ HiGAN-GT 0.687. Counter-intuitively, human-curated boundaries are *least* spatially diverse — real attributes overlap in image space because they share material structure. Spatial diversity is not a quality metric in isolation.

Editing head-to-head (Track 2). [TODO: []Curvature-low pre-selection vs random vs GANSpace-eigenvalue ranking, N=1000 FFHQ test latents, paired ID-cos / CLIP-Δattr / LPIPS at matched target strength. Code: `experiments/baselines/run_editing_head_to_head.py`; ETA ∼30 h.]

Contrastive baselines (Track 5). [TODO: []LatentCLR / DisCo trained from scratch with K=100 directions on FFHQ; will report coverage and curvature ranking through our framework. ETA ∼30 h.]

5.10. Forward vs reverse mode wall-clock

At a fixed accuracy regime on the bedroom generator (RTX 3070 8 GB, fp32, batch 1, 256^2): one first-order saliency map via `torch.func.jvp` takes 38 ms. The same map by per-pixel reverse-mode (vmap+vjp) takes 4.6 s — two orders of magnitude slower because the output dimension $3HW \approx 2\times10^5$ dominates input dimension 1. On FFHQ at 1024^2 the gap widens to $\sim 10^3$. [TODO: []Full benchmark table including A100 numbers in Appendix B.]

6. Application: Compositional Editing Prediction

6.1. Curvature-based pre-selection of editing directions

The C2 ratio $\bar{\rho}_a$ from §5.3 ranks attributes *by what kind of edit they perform* — texture-only edits sit at $\bar{\rho} \approx 1$, topology-changing edits at $\bar{\rho} \gtrsim 20$. Two practical uses follow:

(a) Curvature as an editing-quality filter. Given a candidate pool of N direction discoveries (GANSpace, SeFa, LatentCLR, DisCo, random), filter to the curvature-low subset $\{v : \bar{\rho}_v < \tau\}$ if the goal is clean, identity-preserving edits, or the curvature-high subset if the goal is intentional structural change. Track 2 (experiments/baselines/run_editing_head_to_head.py) implements this on $N = 1000$ FFHQ test latents and reports paired ID-cos / target-Δ-attribute / LPIPS at the StyleSpace equal-strength matching [35]. **TODO**: []Result numbers fill here once Track 2 completes (ETA after Wave 1 queue).]

(b) Predicting compositional editing failure. Using the mixed-Hessian predictor $P(a, b)$ from §5.6, we ask: given two attribute boundaries a, b in a domain we have not yet tested compositionally, can we predict ahead of time whether they will compose linearly or interfere? The honest answer from §5.6 is "yes, *between* curvature regimes; no, *within*". A 28-pair bedroom train / 10-pair FFHQ test hold-out gives a single regime-threshold classifier that identifies structural-vs-textural pair behaviour at $\sim 80\%$ accuracy.

6.2. Unsupervised attribute discovery + CLIP labelling

The C6 pipeline (§5.8, code in `experiments/domains/ffhq/run_discovery_c6.py`) yields candidate attribute boundaries without any human-labelled training data: sample N random unit directions on per-layer W^+ spheres, cluster, label cluster centroids with CLIP. We exploit this in two ways:

Proposing unlabelled directions. On bedroom, the C6 clusters at top-3 CLIP labels recovered 5/8 HiGAN boundaries (precision 1.00). The 3 missed attributes (`wood`, `indoor_lighting`, `cluttered_space`) do not appear in the top-3 of any K=8 cluster, but the corresponding phrases are in the CLIP vocabulary — *the pipeline tells us which attributes the generator's latent geometry doesn't natively stratify*. We propose this as a diagnostic for attribute-taxonomy adequacy.

Recall scaling with N (Track 12). Bedroom recall vs $N \in \{128, 192, 256, 384, 512\}$ tells us how much sampling is sufficient. **TODO**: []fill once Track 12 completes].

6.3. The framework as a cross-method evaluator

The C1 ratio, the C2 path-length / direct-distance metric, the spatial-diversity metric, and the C6 precision/recall pipeline all operate on any K direction vectors regardless of how they were discovered. We use them to evaluate GANSpace, SeFa, LatentCLR, DisCo, random, and the GT InterFaceGAN/HiGAN boundaries on a single comparable axis (§5.9). **Methodologically, this re-frames our paper:** we are not a new direction-discovery method; we are the evaluation framework through which existing methods can be compared, and through which genuinely-novel claims about each method's geometric character (linear vs structural, isotropic vs PCA-like) become measurable.

7. Limitations and Discussion

C4 is a regime indicator, not a within-regime ordering. Our mixed-Hessian predictor $P(a, b)$ has Spearman correlation $+0.48/+0.81/+0.50$ with measured compositional non-linearity across three architectures. However, dropping the highest-curvature attribute from each domain collapses bedroom's Spearman to -0.17 ($n = 21$, $p = 0.47$) and FFHQ's to $+0.60$ ($n = 6$, $p = 0.21$). The single-scalar predictor therefore discriminates *between* structural and textural curvature regimes but does not order pairs within the texture regime. A finer per-pixel predictor that preserves the directional information of $d^2G(b_a, b_b)$ rather than collapsing to its norm is the natural next refinement.

Linear boundaries only. The Lie-bracket analysis in §3 treats attribute directions as constant vector fields on W^+. For learned non-linear edits — e.g. EditGAN's part-conditioned edit vectors [18] or DragGAN's point-based deformation fields [22] — the full Lie bracket of generally varying vector fields is non-zero and may dominate the Hessian term. Extending the curvature prediction to such cases is open.

Diffusion: image-space derivative, not score derivative. For Stable Diffusion we measure $\partial^{1,2}\,\text{decode}(x_0)/\partial(\text{h-space direction})$ via JVP through the DDIM sampling chain, parallel to the StyleGAN measurements. This is strictly different from the score-network Hessian $\nabla_x^2 \log p_t(x)$ studied in [23, 32], which lives in noisy-latent space and concerns the data distribution's local geometry. The two views are complementary: ours is generator-side and operational (predicts edit behaviour), theirs is distribution-side and structural (counts intrinsic dimension). A unified treatment is an attractive next step.

Memory ceiling forces FD-of-JVPs on SD. Composed JVP through the full 50-step DDIM chain at 512^2 OOMs on 8 GB. We therefore compute the second derivative as a finite-difference of two first-order JVPs at $\pm\epsilon$, which keeps peak memory at the single-JVP budget. This is the same construction used by [23] for the diffusion Jacobian. A larger GPU enables direct composed-JVP and would reduce the FD truncation error; we report sensitivity-to-ϵ in Appendix C.

Forward mode is the right tool here, not a panacea. For pushforward and HVP we exploit forward-mode maximally: input dimension 1, output dimension $\sim 10^5$ pixels makes forward 10^3–$10^4\times$ faster than reverse-mode at matched accuracy. For high-dimensional latent-space optimisation (e.g. PTI inversion) reverse mode remains the right choice. We consider forward mode complementary, not a replacement.

Theoretical assumptions. We assume G is C^2 at the sampled latents. ReLU non-smoothness forms a measure-zero set in W^+; we did not encounter numerical issues in practice. A fully rigorous statement requires piece-wise smooth differential geometry — see supplementary remark.

Beyond GANs and diffusion. The differential-geometric view applies, with adaptation, to any generative model defined by a smooth map from a latent space. Direct application to VAEs, normalising flows, and consistency-based generators after suitable smoothing is a clear next step.

Editing claim is descriptive without the head-to-head. The cleanest prescriptive use of our metric — pre-selecting editing directions by curvature regime — requires the editing head-to-head we present in §5.9. In its current form the metric is established as descriptive (curvature correlates with disentanglement, composition, layer localisation); the prescriptive payoff is the empirical question that decides whether the framework changes practice.

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